Hakjoo Oh

Professor · Department of Computer Science and Engineering · College of Informatics, Korea University

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Research Areas

Programming languages, software engineering, and software security.

Education

Ph.D. in Computer Science, Seoul National University — Mar 2007 – Feb 2012
M.S. in Computer Science, Seoul National University — Mar 2005 – Feb 2007
B.S. in Computer Science, KAIST — Mar 2001 – Feb 2005

Employment

Professor, Korea University — Sep 2023 – present
Associate Professor, Korea University — Sep 2018 – Aug 2023
Assistant Professor, Korea University — Mar 2015 – Aug 2018
Research Assistant Professor, Seoul National University — Mar 2012 – Feb 2015
Postdoctoral Researcher, Seoul National University — Mar 2012 – Feb 2014

Awards

2026 **Amazon Research Award**, Amazon Security call, Spring 2026 cycle
Verifix: Towards Verifiable Security Patching with Neuro-Symbolic Agents

2026 **ACM SIGSOFT Distinguished Paper & Artifact Awards**, FSE 2026
Pig: Leveraging Large Language Models for Python Library Migrations

2026 **ACM SIGSOFT Distinguished Artifact Award**, ICSE 2026
TypeCare: Boosting Python Type Inference Models via Context-Aware Re-Ranking and Augmentation

2022 **ACM SIGSOFT Distinguished Paper Award**, ICSE 2022
SymTuner: Maximizing the Power of Symbolic Execution by Adaptively Tuning External Parameters

2020 **ACM SIGSOFT Distinguished Paper Award**, ISSTA 2020
Effective White-box Testing of Deep Neural Networks with Adaptive Neuron-Selection Strategy

2019 **ACM SIGSOFT Distinguished Paper Award**, ICSE 2019
Resource-aware Program Analysis via Online Abstraction Coarsening

2016 **Best Paper Award**, GPCE 2016
Synthesizing Regular Expressions from Examples for Introductory Automata Assignments

Publications

Reverse chronological order. * Equal contribution.

78 **ICSE** 2027 **Prompt Less, Infer More: Static Analysis-Guided LLM Type Annotation for Python**
Wonseok Oh, Michael Pradel, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

77 **ISSTA Demo** 2026 **Fusion: A Fold and Unfold Annotation Synthesis Tool for Viper-Based Verification**
Joonghoon Lee, Nicolas Klose, Seokhyun Lee, Peter Müller, and **Hakjoo Oh**
ACM SIGSOFT International Symposium on Software Testing and Analysis, Tool Demonstrations

76 **ISSTA Demo** 2026
SCOUT: A Scenario-Guided Unit Test Generation Tool for Compact Open-Source LLMs
Myungho Lee, Seokhyeon Moon, Yoon-Chan Jhi, and **Hakjoo Oh**
ACM SIGSOFT International Symposium on Software Testing and Analysis, Tool Demonstrations

75 **ASE Industry** 2026
EvoLLM: High-Coverage, Readable, and Token-Efficient Unit Test Generation for Industrial Java Codebases
Seokhyeon Moon, Myungho Lee, Jihun Kim, Yoon-Chan Jhi, Jihoon Cho, and **Hakjoo Oh**
IEEE/ACM International Conference on Automated Software Engineering, Industry Showcase Track

74 **OOPSLA** 2026
Prunario: Testing Autonomous Driving Systems by Pruning Likely Redundant Scenarios
Minsu Kim, Sunbeom So, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

73 **FSE** 2026
Reducing Coverage-Equivalent Inputs in Grammar-based Fuzzing by Avoiding Recurrent Rule Sequences
Jaehan Yoon, Yunji Seo, **Hakjoo Oh**, and Sooyoung Cha
ACM International Conference on the Foundations of Software Engineering

72 **FSE** 2026
Pig: Leveraging Large Language Models for Python Library Migrations
Miryeong Kang, Wonseok Oh, Gabin An, and **Hakjoo Oh**
ACM International Conference on the Foundations of Software Engineering
ACM SIGSOFT Distinguished Paper & Artifact Awards

71 **ICSE Demo** 2026
PySTAAR: An End-to-End, Extensible Framework for Automated Python Type Error Repair
Wonseok Oh, Hyobin Park, Miryeong Kang, Seungbin Choi, Yunja Choi, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering, Demonstrations Track

70 **ICSE** 2026
TypeCare: Boosting Python Type Inference Models via Context-Aware Re-Ranking and Augmentation
Wonseok Oh and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering
ACM SIGSOFT Distinguished Artifact Award

69 **OOPSLA** 2025
Enhancing APR with PRISM: A Semantic-Based Approach to Overfitting Patch Detection
Dowon Song and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

68 **OOPSLA** 2025
PAFL: Enhancing Fault Localizers by Leveraging Project-Specific Fault Patterns
Donguk Kim, Minseok Jeon, Doha Hwang, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

67 **ASE** 2024
Towards Effective Static Type-Error Detection for Python
Wonseok Oh and **Hakjoo Oh**
IEEE/ACM International Conference on Automated Software Engineering

66 **ASE** 2024
Effective Unit Test Generation for Java Null Pointer Exceptions
Myungho Lee, Jiseong Bak, Seokhyeon Moon, Yoon-Chan Jhi, and **Hakjoo Oh**
IEEE/ACM International Conference on Automated Software Engineering

65 **OOPSLA** 2024
Quantum Probabilistic Model Checking for Time-Bounded Properties
Seungmin Jeon, Kyeongmin Cho, Chan Gu Kang, Janggun Lee, **Hakjoo Oh**, and Jeehoon Kang
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

64 **OOPSLA** 2024
Statistical Testing of Quantum Programs via Fixed-Point Amplitude Amplification
Chan Gu Kang, Joonghoon Lee, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

63 **PLDI** 2024
PL4XGL: A Programming Language Approach to Explainable Graph Learning
Minseok Jeon, Jihyeok Park, and **Hakjoo Oh**
ACM SIGPLAN Conference on Programming Language Design and Implementation

62 **ESEC/FSE** 2023
SmartFix: Fixing Vulnerable Smart Contracts by Accelerating Generate-and-Verify Repair using Statistical Models

Sunbeom So and **Hakjoo Oh**
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

61 **USENIX** 2023
V1SCAN: Discovering 1-day Vulnerabilities in Reused C/C++ Open-source Software Components Using Code Classification Techniques
Seunghoon Woo, Eunjin Choi, Heejo Lee, and **Hakjoo Oh**
USENIX Security Symposium

60 **TOPLAS** 2023
Optimizing Homomorphic Evaluation Circuits by Program Synthesis and Time-Bounded Exhaustive Search
Dongkwon Lee, Woosuk Lee, **Hakjoo Oh**, and Kwangkeun Yi
ACM Transactions on Programming Languages and Systems

59 **OOPSLA** 2023
Modular Component-based Quantum Circuit Synthesis
Chan Gu Kang and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

58 **ICST Industry** 2023
Automating Endurance Test for Flash-based Storage Devices in Samsung Electronics
Jinkook Kim, Minseok Jeon, Sejeong Jang, and **Hakjoo Oh**
IEEE International Conference on Software Testing, Verification and Validation, Industry Track

57 **ICSE** 2023
Learning Seed-Adaptive Mutation Strategies for Greybox Fuzzing
Myungho Lee, Sooyoung Cha, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

56 **ICSE** 2023
Diver: Oracle-Guided SMT Solver Testing with Unrestricted Random Mutations
Jongwook Kim, Sunbeom So, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

55 **ICSE** 2023
Learning to Boost Disjunctive Static Bug-Finders
Yoonseok Ko and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

54 **ESEC/FSE** 2022
PyTER: Effective Program Repair for Python Type Errors
Wonseok Oh and **Hakjoo Oh**
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

53 **ICSE** 2022
SymTuner: Maximizing the Power of Symbolic Execution by Adaptively Tuning External Parameters
Sooyoung Cha, Myungho Lee, Seokhyun Lee, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering
ACM SIGSOFT Distinguished Paper Award

52 **ICSE** 2022
NPEX: Repairing Java Null Pointer Exceptions without Tests
Junhee Lee*, Seongjoon Hong*, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

51 **POPL** 2022
Return of CFA: Call-Site Sensitivity Can Be Superior to Object Sensitivity Even for Object-Oriented Programs
Minseok Jeon and **Hakjoo Oh**
ACM SIGPLAN Symposium on Principles of Programming Languages

50 **TSE** 2021
Enhancing Dynamic Symbolic Execution by Automatically Learning Search Heuristics
Sooyoung Cha, Seongjoon Hong, Jiseong Bak, Jingyoung Kim, Junhee Lee, and **Hakjoo Oh**
IEEE Transactions on Software Engineering (accepted)

49 **ESEC/FSE** 2021
Context-Aware and Data-Driven Feedback Generation for Programming Assignments
Dowon Song, Woosuk Lee, and **Hakjoo Oh**
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

48 **IST** 2021
A Practical Algorithm for Learning Disjunctive Abstraction Heuristics in Static Program Analysis
Donghoon Jeon, Minseok Jeon, and **Hakjoo Oh**
Information and Software Technology

47 **USENIX** 2021
SmarTest: Effectively Hunting Vulnerable Transaction Sequences in Smart Contracts through Language Model-Guided Symbolic Execution
Sunbeom So, Seongjoon Hong, and **Hakjoo Oh**
USENIX Security Symposium

46 **ICSE** 2021
CENTRIS: A Precise and Scalable Approach for Identifying Modified Open-Source Software Reuse
Seunghoon Woo, Sunghan Park, Seulbae Kim, Heejo Lee, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

45 **OOPSLA** 2020
Learning Graph-based Heuristics for Pointer Analysis without Handcrafting Application-Specific Features
Minseok Jeon, Myungho Lee, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

44 **ESEC/FSE** 2020
Making Symbolic Execution Promising by Learning Aggressive State-Pruning Strategy
Sooyoung Cha and **Hakjoo Oh**
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

43 **ISSTA** 2020
Effective White-box Testing of Deep Neural Networks with Adaptive Neuron-Selection Strategy
Seokhyun Lee, Sooyoung Cha, Dain Lee, and **Hakjoo Oh**
ACM SIGSOFT International Symposium on Software Testing and Analysis
ACM SIGSOFT Distinguished Paper Award

42 **PLDI** 2020
Optimizing Homomorphic Evaluation Circuits by Program Synthesis and Term Rewriting
Dongkwon Lee, Woosuk Lee, **Hakjoo Oh**, and Kwangkeun Yi
ACM SIGPLAN Conference on Programming Language Design and Implementation

41 **ICSE** 2020
SAVER: Scalable, Precise, and Safe Memory-Error Repair
Seongjoon Hong*, Junhee Lee*, Jeongsoo Lee, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

40 **S&P** 2020
VeriSmart: A Highly Precise Safety Verifier for Ethereum Smart Contracts
Sunbeom So, Myungho Lee, Jisu Park, Heejo Lee, and **Hakjoo Oh**
IEEE Symposium on Security and Privacy

39 **OOPSLA** 2019
Automatic and Scalable Detection of Logical Errors in Functional Programming Assignments
Dowon Song, Myungho Lee, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

38 **ESEC/FSE** 2019
Concolic Testing with Adaptively Changing Search Heuristics
Sooyoung Cha and **Hakjoo Oh**
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

37 **ICSE** 2019
Resource-aware Program Analysis via Online Abstraction Coarsening
Kihong Heo, **Hakjoo Oh**, and Hongseok Yang
IEEE/ACM International Conference on Software Engineering
ACM SIGSOFT Distinguished Paper Award

36 **TOPLAS** 2019
A Machine-Learning Algorithm with Disjunctive Model for Data-Driven Program Analysis
Minseok Jeon*, Sehun Jeong*, Sungdeok Cha, and **Hakjoo Oh**
ACM Transactions on Programming Languages and Systems

35 **ASE** 2018
Template-Guided Concolic Testing via Online Learning

Sooyoung Cha, Seonho Lee, and **Hakjoo Oh**
IEEE/ACM International Conference on Automated Software Engineering

34 **IST** 2018
A Scalable Learning Algorithm for Data-Driven Program Analysis
Sooyoung Cha, Sehun Jeong, and **Hakjoo Oh**
Information and Software Technology

33 **OOPSLA** 2018
Precise and Scalable Points-to Analysis via Data-Driven Context Tunneling
Minseok Jeon, Sehun Jeong, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

32 **OOPSLA** 2018
Automatic Diagnosis and Correction of Logical Errors for Functional Programming Assignments
Junho Lee, Dowon Song, Sunbeom So, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

31 **ESEC/FSE** 2018
MemFix: Static Analysis-Based Repair of Memory Deallocation Errors for C
Junhee Lee*, Seongjoon Hong*, and **Hakjoo Oh**
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

30 **IJCAI** 2018
Synthesizing Pattern Programs from Examples
Sunbeom So and **Hakjoo Oh**
International Joint Conference on Artificial Intelligence

29 **ICSE** 2018
Automatically Generating Search Heuristics for Concolic Testing
Sooyoung Cha, Seongjoon Hong, Junhee Lee, and **Hakjoo Oh**
IEEE/ACM International Conference on Software Engineering

28 **TOPLAS** 2018
Adaptive Static Analysis via Learning with Bayesian Optimization
Kihong Heo, **Hakjoo Oh**, Hongseok Yang, and Kwangkeun Yi
ACM Transactions on Programming Languages and Systems

27 **FMSD** 2018
Learning Analysis Strategies for Octagon and Context Sensitivity from Labeled Data Generated by Static Analyses
Kihong Heo, **Hakjoo Oh**, and Hongseok Yang
Formal Methods in System Design (invited paper)

26 **TOPLAS** 2017
Sound Non-Statistical Clustering of Static Analysis Alarms
Woosuk Lee, Wonchan Lee, Dongok Kang, Kihong Heo, **Hakjoo Oh**, and Kwangkeun Yi
ACM Transactions on Programming Languages and Systems

25 **OOPSLA** 2017
Data-Driven Context-Sensitivity for Points-to Analysis
Sehun Jeong*, Minseok Jeon*, Sungdeok Cha, and **Hakjoo Oh**
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

24 **OOPSLA** 2017
Automatically Generating Features for Learning Program Analysis Heuristics for C-like Languages
Kwonsoo Chae, **Hakjoo Oh**, Kihong Heo, and Hongseok Yang
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

23 **SAS** 2017
Synthesizing Imperative Programs from Examples Guided by Static Analysis
Sunbeom So and **Hakjoo Oh**
Static Analysis Symposium

22 **IJCAI** 2017
End-to-End Prediction of Buffer Overruns from Raw Source Code via Neural Memory Networks
Min-je Choi, Sehun Jeong, **Hakjoo Oh**, and Jaegul Choo
International Joint Conference on Artificial Intelligence

21 **S&P** 2017
VUDDY: A Scalable Approach for Vulnerable Code Clone Discovery
Seulbae Kim, Seunghoon Woo, Heejo Lee, and **Hakjoo Oh**
IEEE Symposium on Security and Privacy

20 **ICSE** 2017
Machine-Learning-Guided Selectively Unsound Static Analysis
Kihong Heo, **Hakjoo Oh**, and Kwangkeun Yi
IEEE/ACM International Conference on Software Engineering

19 **SP&E** 2017
Selective Conjunction of Context-Sensitivity and Octagon Domain toward Scalable and Precise Global Static Analysis
Kihong Heo, **Hakjoo Oh**, and Kwangkeun Yi
Software: Practice and Experience

18 **TOPLAS** 2016
Selective X-Sensitive Analysis Guided by Impact Pre-Analysis
Hakjoo Oh, Wonchan Lee, Kihong Heo, Hongseok Yang, and Kwangkeun Yi
ACM Transactions on Programming Languages and Systems, 38(2)

17 **SAS** 2016
Learning a Variable-Clustering Strategy for Octagon from Labeled Data Generated by a Static Analysis
Kihong Heo, **Hakjoo Oh**, and Hongseok Yang
Static Analysis Symposium

16 **GPCE** 2016
Synthesizing Regular Expressions from Examples for Introductory Automata Assignments
Mina Lee*, Sunbeom So*, and **Hakjoo Oh**
ACM SIGPLAN International Conference on Generative Programming: Concepts and Experiences
Best Paper Award

15 **APLAS** 2016
Learning a Strategy for Choosing Widening Thresholds from a Large Codebase
Sooyoung Cha, Sehun Jeong, and **Hakjoo Oh**
Asian Symposium on Programming Languages and Systems

14 **SP&E** 2016
Widening with Thresholds via Binary Search
Sol Kim, Kihong Heo, **Hakjoo Oh**, and Kwangkeun Yi
Software: Practice and Experience

13 **IFIP SEC** 2016
Automated Source Code Instrumentation for Verifying Potential Vulnerabilities
Hongzhe Li, Jaesang Oh, **Hakjoo Oh**, and Heejo Lee
IFIP International Information Security and Privacy Conference

12 **OOPSLA** 2015
Learning a Strategy for Adapting a Program Analysis via Bayesian Optimisation
Hakjoo Oh, Hongseok Yang, and Kwangkeun Yi
ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications

11 **PLDI** 2014
Selective Context-Sensitivity Guided by Impact Pre-Analysis
Hakjoo Oh, Wonchan Lee, Kihong Heo, Hongseok Yang, and Kwangkeun Yi
ACM SIGPLAN Conference on Programming Language Design and Implementation

10 **TOPLAS** 2014
Global Sparse Analysis Framework
Hakjoo Oh, Kihong Heo, Wonchan Lee, Woosuk Lee, Daejun Park, Jeehoon Kang, and Kwangkeun Yi
ACM Transactions on Programming Languages and Systems, 36(3)

9 **SAS** 2014
A Progress Bar for Static Analyzers
Woosuk Lee, **Hakjoo Oh**, and Kwangkeun Yi
Static Analysis Symposium

8 **COMLAN** 2014
A Sparse Evaluation Technique for Detailed Semantic Analyses
Yoonseok Ko, Kihong Heo, and **Hakjoo Oh**
Computer Languages, Systems and Structures, 40(3–4)

7 **SCP** 2013
Access-based Abstract Memory Localization in Static Analysis
Hakjoo Oh and Kwangkeun Yi
Science of Computer Programming, 78(9):1701–1727

6 **PLDI** 2012
Design and Implementation of Sparse Global Analyses for C-like Languages
Hakjoo Oh, Kihong Heo, Wonchan Lee, Woosuk Lee, and Kwangkeun Yi
ACM SIGPLAN Conference on Programming Language Design and Implementation

First PLDI paper from Korea

5 **APLAS** 2011
Access-based Localization with Bypassing
Hakjoo Oh and Kwangkeun Yi
Asian Symposium on Programming Languages and Systems

4 **VMCAI** 2011
Access-analysis-based Tight Localization of Abstract Memories
Hakjoo Oh, Lucas Brutschy, and Kwangkeun Yi
International Conference on Verification, Model Checking, and Abstract Interpretation

3 **SP&E** 2010
An Algorithmic Mitigation of Large Spurious Interprocedural Cycles in Static Analysis
Hakjoo Oh and Kwangkeun Yi
Software: Practice and Experience, 40(8):585–603

2 **APLAS** 2009
Large Spurious Cycles in Global Static Analysis and Its Algorithmic Mitigation
Hakjoo Oh
Asian Symposium on Programming Languages and Systems

1 **DSMM** 2009
Identifying Static Analysis Techniques for Finding Non-fix Hunks in Fix Revisions
Yungbum Jung, **Hakjoo Oh**, and Kwangkeun Yi
ACM Workshop on Data-intensive Software Management and Mining

Academic Activities

Memberships

Member, IFIP Working Group 2.4 (Software Implementation Technology) ifip-wg24.org 2025 – present

Program Committees

2027 OOPSLA (Associate Chair), FSE
2026 SAS, ASE, ISSTA, PLDI
2025 SAS (PC Co-Chair), ASE, ICFP, OOPSLA, POPL, ICSE
2024 PLDI, ISSTA, ICSE
2023 ISSTA, CC
2022 OOPSLA, ISSTA, PLDI
2021 APLAS (PC Chair), OOPSLA, OCaml Workshop, WoSCA, ECOOP, ICSE
2020 TAPAS (PC Co-Chair), APLAS, WoSCA, OOPSLA (External Review Committee), LCTES, ICSE-SEIP, ATVA
2019 SAS (Artifact Evaluation Chair), CAV, ATVA
2018 APLAS, SAS, OOPSLA
2017 APLAS
2015 APLAS, SAC (Programming Languages Track)
2014 SAC (Programming Languages Track)
2013 APLAS